Math340Fa23-Midterm1

Thursday, October 5, 2023 2:41 PM

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Math 340: Elementary Matrix and Linear Algebra

MIDTERM ONE

Wednesday, October 11th, 2023, 7:30pm-9:00pm

Please circle your Instructor and your Discussion Session (date and start time given) in the table below.

Dr. Kaitlyn Phillipson	Dr. Shengwen Gan	Dr. Joshua Mundinger	Dr. Feng Zhu
Joey (Yu), T 3:30 PM	Zaidan, M 12:05 PM	Paulius, M 7:45 AM	Anna, T 1:20 PM
Joey (Yu), T 4:35 PM	Zaidan, W 12:05 PM	Paulius, M 8:50 AM	Anna, T 2:25 PM
Joey (Yu), R 3:30 PM	Daniel, M 12:05 PM	Paulius, W 7:45 AM	Anna, R 2:25 PM
Yaoxian, R 4:35 PM	Daniel, W 1:20 PM	Jing, W 8:50 AM	
Hongyu, R 9:55 AM	Dewei, M 3:30 PM	Jiaqi, M 11:00 AM	
Hongyu, T 9:55 AM	Dewei, W 3:30 PM	Jiaqi, W 12:05 PM	
Hongyu, R 8:50 AM	Dewei, M 2:25 PM	Ellie, W 1:20 PM	
	Joel, W 4:35 PM	Ellie, M 1:20 PM	

Name: Pepe Silvia Wisc email: ______________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______________________

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of nine problems for a total of 50 points; the exam is 7 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (5 points, 1 point each) Are each of the following true or false statements? Fill in the bubble next to the correct answer. Justification is not necessary.

a.) If $A\mathbf{x} = \mathbf{b}$ is consistent, then $\mathbf{b}$ is a linear combination of the columns of A.

- ● True
- ○ False

$A\vec{x}$ is literally a linear combination of the columns of A.

b.) If A is a 3×3 matrix and $\det(A) = 2$, then $\det(3A) = 6$. X

- ○ True
- ● False

$3^3 \cdot 2$

c.) If $A, B,$ and C are 3×3 invertible (also called nonsingular) matrices, and $C = AB^{-1}$, then $C^{-1} = BA^{-1}$.

- ● True
- ○ False

$$C^{-1} = (AB^{-1})^{-1} = (B^{-1})^{-1}A^{-1} = BA^{-1}$$

d.) If A is a 4×4 matrix that is row equivalent to I_4, then $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

- ○ True
- ● False

$[A \,\vdots\, \vec{0}] \longrightarrow [I_4 \,\vdots\, \vec{0}]$ unique (trivial) solution.

e.) If $T : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix transformation given by $T(\mathbf{x}) = A\mathbf{x}$, and A is a 2×3 matrix, then $n = 2$ and $m = 3$.

- ○ True
- ● False

$$\underbrace{A}_{2\times 3}\underbrace{\vec{x}}_{3\times 1} \to 2\times 1$$

$\vec{x}$ is $3\times 1 \to \mathbb{R}^3$ $n=3$

$A\vec{x}$ is $2\times 1 \to \mathbb{R}^2$ $m=2$

2. (3 points) Find the eigenvalues of A, where $A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$. Show all work.

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda-2 & -3 \\ -5 & \lambda-4 \end{bmatrix} = (\lambda-2)(\lambda-4) - 15$$

$$= \lambda^2 - 6\lambda + 8 - 15$$

$$= \lambda^2 - 6\lambda - 7$$

$$= (\lambda-7)(\lambda+1)$$

$$\boxed{\lambda = -1, 7}$$

3. (6 points total) Let $T : \mathbb{R}^2 \to \mathbb{R}^4$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ 3x_2 - x_1 \\ 2x_1 - x_2 \\ 3x_2 \end{bmatrix}$. You can assume that T is a linear transformation.

a. (3 points) Find the (standard) matrix A such that $T(\mathbf{x}) = A\mathbf{x}$. Show all work.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \end{bmatrix} \qquad T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \\ -1 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 3 \\ 2 & -1 \\ 0 & 3 \end{bmatrix}$$

b. (3 points) Is there an $\mathbf{x}$ in $\mathbb{R}^2$ such that $T(\mathbf{x}) = \begin{bmatrix} 3 \\ -2 \\ 3 \\ 3 \end{bmatrix}$? Fully justify your answer.

Need: $x_1 + x_2 = 3$ $\quad x_1 = 2$ Need

$3x_2 - x_1 = -2$

$2x_1 - x_2 = 3$

$3x_2 = 3 \Rightarrow x_2 = 1$ Need

$2(2) - 1 = 3 \checkmark.$

$3(1) - 2 = 1 \neq -2.$

No, no such $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ exists (see work above)

4. (5 points) Let $A = \begin{bmatrix} 2 & x \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$. Find x (if possible) so that $AB + C$ is a **symmetric matrix**. Show your work.

$$AB = \begin{bmatrix} 2 & x \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -2+2x \\ 0 & -3+8 \end{bmatrix}$$

$$AB + C = \begin{bmatrix} 0 & -2+2x \\ 0 & 5 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4+2x \\ 3 & 6 \end{bmatrix}$$

Need: $3 = -4 + 2x$

$7 = 2x \quad \boxed{x = 7/2}$

5. (7 points total)

Let A, B, C, D be 4×4 matrices with $\det(A) = 4$, $\det(B) = -3$, $\det(C) = -2$, $\det(D) = 0$.

a. (3 points) Find $\det(C^{-1}B^T A^2)$. Show all work.

$$\overset{\shortparallel}{\frac{1}{\det(C)}} \det(B) (\det(A))^2 = \frac{1}{-2} \cdot (-3) \cdot 16$$

$$= 8(3) = \boxed{24}$$

b. (2 points) Is AD invertible? Justify your answer in one sentence.

$\det(AD) = \det(A)\det(D) = 4 \cdot 0 = 0.$

No, the determinant $= 0$, so AD is <u>not</u> invertible.

c. (2 points) Suppose matrix M is obtained from A by the row operation $10r_1 + r_2 \to r_1$ (hint: **read this row operation very carefully**). What is $\det(M)$? Justify your answer.

This is <u>not</u> an elementary row operation.

This is: $10r_1 \to r_1$ (mult. det by 10)

<u>Then</u> $r_1 + r_2 \to r_1$ (no change)

so $\det(M) = 40$

6. (6 points total) (This problem has overflow room on the next page if needed)

Let $C = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -1 & -2 \\ 2 & -2 & -3 \end{bmatrix}$.

a. (4 points) Compute C^{-1}. Show all work.

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & -2 & 0 & 1 & 0 \\ 2 & -2 & -3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & -2 & -3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{-2r_1 + r_3 \to r_3} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 1 \end{array}\right] \xrightarrow{2r_3 + r_1 \to r_1} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & -3 & 2 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{r_2 + r_1 \to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -3 & 2 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 1 \end{array}\right] \qquad C^{-1} = \begin{bmatrix} 1 & -3 & 2 \\ 1 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

b. (2 points) Solve $C\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$. Show all work (hint: you can use your answer to part (a)).

$$\vec{x} = \begin{bmatrix} 1 & -3 & 2 \\ 1 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$$

(Overflow room for Problem 6, if needed)

7. (4 points) Suppose $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 3 & 1 \\ -1 & 2 & x \end{bmatrix}$.

If the rank of A is 2, find x. Show all work, and justify your answer.

Soln1:

Reduce...

$$A \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 4 & 3 & 1 \\ 3 & 2 & 1 \\ -1 & 2 & x \end{bmatrix} \xrightarrow{-r_2 + r_1 \to r_1} \begin{bmatrix} 1 & 1 & 0 \\ 3 & 2 & 1 \\ -1 & 2 & x \end{bmatrix} \xrightarrow[r_1 + r_3 \to r_3]{-3r_1 + r_2 \to r_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 3 & x \end{bmatrix}$$

$$\xrightarrow{3r_2 + r_3 \to r_3} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & x+3 \end{bmatrix} \longrightarrow \text{Need: } x + 3 = 0 \Rightarrow \boxed{x = -3}$$

Soln2: $\det(A) = 0$ b/c A is not invertible $(\text{rank}(A) = 2 < 3)$

$$\det(A) = \det \begin{bmatrix} 3 & 2 & 1 \\ 4 & 3 & 1 \\ -1 & 2 & x \end{bmatrix} = 3(3x-2) - 2(4x+1) + 1(8+3)$$

$$= 9x - 6 - 8x - 2 + 11 = x + 3 = 0$$

$$\boxed{x = -3}$$

8. (7 points total) Suppose a linear system of 3 equations and 3 unknowns (x, y, z) reduces to the following form:

$$\left[\begin{array}{ccc:c} 1 & 0 & 4 & 3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & (a-3) & (a^2-9) \end{array}\right]$$

a. (2 points) For which value(s) of a (if any) does this system have **no solutions**? Briefly explain your answer.

None, b/c if $a \neq 3 \Rightarrow$ we can get $[0\ 0\ 1 \mid a+3]$ for last row $\Rightarrow$ unique soln.
If $a = 3 \Rightarrow$ last row is $[0\ 0\ 0 \mid 0] \to$ inf. many solutions.

b. (2 points) For which value(s) of a (if any) does this system have **infinitely many solutions**? Briefly explain your answer.

If $a = 3$, then we have a consistent system w/ z a free variable $\Rightarrow$ inf. many solutions.

c. (3 points) Using $a = 1$, finish solving the system. If it has infinitely many solutions, write the solution as a parameterized set.

$$\left[\begin{array}{ccc:c} 1 & 0 & 4 & 3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & -2 & -8 \end{array}\right] \xrightarrow{-\frac{1}{2}r_3 \to r_3} \left[\begin{array}{ccc:c} 1 & 0 & 4 & 3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow{2r_3 + r_2 \to r_2} \left[\begin{array}{ccc:c} 1 & 0 & 4 & 3 \\ 0 & 1 & 0 & 13 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

$$\xrightarrow{-4r_3 + r_1 \to r_1} \left[\begin{array}{ccc:c} 1 & 0 & 0 & -13 \\ 0 & 1 & 0 & 13 \\ 0 & 0 & 1 & 4 \end{array}\right] \quad \boxed{\begin{array}{l} x = -13 \\ y = 13 \\ z = 4 \end{array}}$$

9. (7 points total) (This problem has overflow room on the next page if needed)

Suppose A is a 4×4 matrix, and $\mathbf{u} = \begin{bmatrix}1\\-2\\4\\3\end{bmatrix}$ is an eigenvector of A with eigenvalue 2, and $\mathbf{v} = \begin{bmatrix}2\\0\\1\\0\end{bmatrix}$ is an eigenvector of A with eigenvalue 3.

Compute each of the following. Show all work.

a. (1 point)

$$A\begin{bmatrix}1\\-2\\4\\3\end{bmatrix} = 2\begin{bmatrix}1\\-2\\4\\3\end{bmatrix} = \begin{bmatrix}2\\-4\\8\\6\end{bmatrix}$$

b. (2 points)

$$A\begin{bmatrix}-4\\0\\-2\\0\end{bmatrix} = A(-2)\begin{bmatrix}2\\0\\1\\0\end{bmatrix} = -2A\begin{bmatrix}2\\0\\1\\0\end{bmatrix} = -2\begin{bmatrix}6\\0\\3\\0\end{bmatrix} = \begin{bmatrix}-12\\0\\-6\\0\end{bmatrix}$$

c. (4 points)

$$A\begin{bmatrix}3\\2\\-2\\-3\end{bmatrix} = A\left((-1)\begin{bmatrix}1\\-2\\4\\3\end{bmatrix} + 2\begin{bmatrix}2\\0\\1\\0\end{bmatrix}\right) = -1\begin{bmatrix}2\\-4\\8\\6\end{bmatrix} + 2\begin{bmatrix}6\\0\\3\\0\end{bmatrix} = \begin{bmatrix}10\\4\\-2\\-6\end{bmatrix}$$

(hint: Is this vector a linear combination of $\mathbf{u}$ and $\mathbf{v}$?)

$$\begin{bmatrix}3\\2\\-2\\-3\end{bmatrix} = a\begin{bmatrix}1\\-2\\4\\3\end{bmatrix} + b\begin{bmatrix}2\\0\\1\\0\end{bmatrix} = \begin{bmatrix}a+2b\\-2a\\4a+b\\3a\end{bmatrix}$$

$a+2b=3 \rightsquigarrow b=\frac{3+1}{2}=2$

$-2a=2 \quad a=-1$

$4(-1)+2=-2$ ✓

$3a=-3$

(Overflow room for Problem 9, if needed)

Scratch paper page!!

Back of scratch paper page!